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MLDL

Experiment No 03

Aim: Application of Decision Tree and Random Forest for Classification

1. Dataset Source

The dataset used for this experiment is the **Fetal Health Classification Dataset**, which is publicly available on Kaggle.

Dataset Link:
https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification

This dataset is widely used in healthcare analytics to classify fetal health conditions based on cardiotocography (CTG) exam results.

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2. Dataset Description

The Fetal Health dataset contains medical measurements obtained from cardiotocography exams, which monitor fetal heart rate and uterine contractions during pregnancy. The goal of this experiment is to classify the **health condition of the fetus**.

Dataset Characteristics

- Number of instances: 2126 records
- Type: Structured, numerical dataset
- Problem Type: Multi-class classification
- Domain: Healthcare / Medical Diagnosis

Features Used

The dataset consists of several numerical attributes related to fetal heart rate patterns, such as:

- Baseline fetal heart rate
- Accelerations
- Fetal movements
- Uterine contractions
- Light, severe, and prolonged decelerations

- Variability measures

These features help in identifying abnormal fetal conditions.

Target Variable

- **fetal_health**
  - 1 → Normal
  - 2 → Suspect
  - 3 → Pathological

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3. Mathematical Formulation of the Algorithms

Decision Tree Classifier

A Decision Tree classifier predicts the class label by learning a set of decision rules derived from input features. The dataset is recursively split into subsets based on feature values.

To select the best split, impurity measures such as **Gini Index** or **Entropy** are used.

Gini Index:

$$Gini = 1 - \Sigma(p_i)^2$$

Where:

- p_i is the probability of class i at a node

The tree continues splitting until a stopping condition is reached.

Random Forest Classifier

Random Forest is an ensemble learning technique that builds multiple decision trees using random subsets of data and features.

Final Prediction:
Majority vote of predictions from all decision trees

By combining multiple trees, Random Forest reduces overfitting and improves classification accuracy.

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4. Algorithm Limitations

Decision Tree Limitations

- Prone to overfitting, especially with deep trees
- Sensitive to noise and small variations in data
- Can become complex and difficult to interpret

Random Forest Limitations

- Requires more computational power and memory
- Less interpretable compared to a single decision tree
- Training time increases with the number of trees

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5. Methodology / Workflow

Step-by-Step Procedure

1. Dataset loading using Pandas
2. Separation of input features and target variable
3. Splitting the dataset into training (80%) and testing (20%) sets
4. Training a Decision Tree classifier
5. Training a Random Forest classifier
6. Predicting fetal health classes on test data
7. Evaluating models using classification metrics
8. Visualizing decision tree structure and feature importance

Workflow Diagram (Textual Representation)

Dataset → Preprocessing → Train-Test Split
→ Model Training → Prediction
→ Evaluation → Visualization

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6. Performance Analysis

The performance of the classification models was evaluated using the following metrics:

- **Accuracy Score:** Measures overall correctness of predictions

- **Confusion Matrix:** Shows correct and incorrect predictions for each class
- **Classification Report:** Includes precision, recall, and F1-score

Observations

- Decision Tree provides reasonable accuracy but may overfit the training data
- Random Forest achieves higher accuracy due to ensemble learning
- Random Forest shows better generalization on unseen data

Overall, Random Forest outperforms Decision Tree in terms of stability and predictive performance.

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7. Hyperparameter Tuning

Decision Tree Hyperparameters

- **max_depth:** Controls the depth of the tree
- **min_samples_split:** Minimum samples required to split a node

Random Forest Hyperparameters

- **n_estimators:** Number of trees in the forest
- **max_depth:** Maximum depth of each tree
- **max_features:** Number of features considered at each split

Tuning Impact

- Increasing tree depth improves accuracy but increases overfitting risk
- Increasing number of trees improves Random Forest performance but increases computation time
- Proper tuning improves model accuracy and generalization

CODE:

```
# ==============================
# IMPORT LIBRARIES
# ==============================
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
```

```python
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report


# ================================
# LOAD DATASET
# ================================
df = pd.read_csv("fetal_health.csv")

print(df.head())
print(df.shape)
print(df['fetal_health'].value_counts())


# ================================
# FEATURES & TARGET
# ================================
X = df.drop("fetal_health", axis=1)
y = df["fetal_health"]   # 1 = Normal, 2 = Suspect, 3 = Pathological


# ================================
# TRAIN-TEST SPLIT
# ================================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# ================================
# DECISION TREE CLASSIFIER
# ================================
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)

print("\n--- Decision Tree Classifier ---")
```

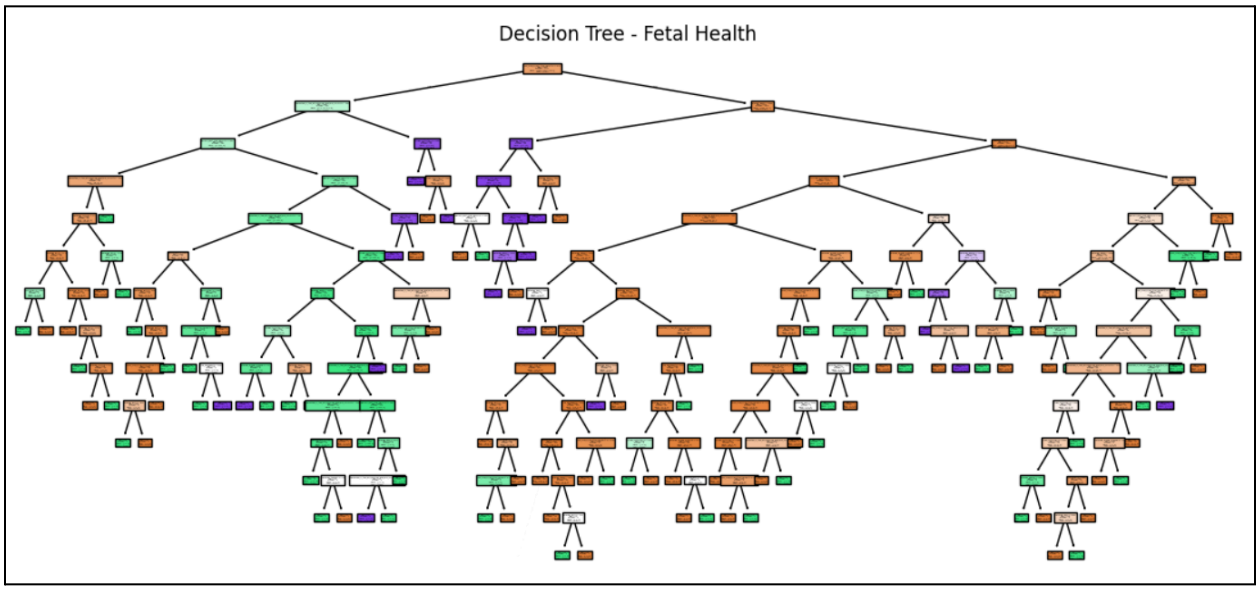
```
print("Accuracy:", accuracy_score(y_test, y_pred_dt))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_dt))
print("Classification Report:\n", classification_report(y_test,
y_pred_dt))


# ================================
# DECISION TREE VISUALIZATION
# ================================
plt.figure(figsize=(14, 6))
plot_tree(
    dt,
    feature_names=X.columns,
    class_names=["Normal", "Suspect", "Pathological"],
    filled=True
)
plt.title("Decision Tree - Fetal Health")
plt.show()


# ================================
# RANDOM FOREST CLASSIFIER
# ================================
rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

print("\n--- Random Forest Classifier ---")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_rf))
print("Classification Report:\n", classification_report(y_test,
y_pred_rf))


# ================================
# FEATURE IMPORTANCE (RANDOM FOREST)
```

```python
# ================================
importances = rf.feature_importances_

plt.figure(figsize=(8, 6))
plt.barh(X.columns, importances)
plt.xlabel("Importance")
plt.title("Random Forest Feature Importance")
plt.show()
```

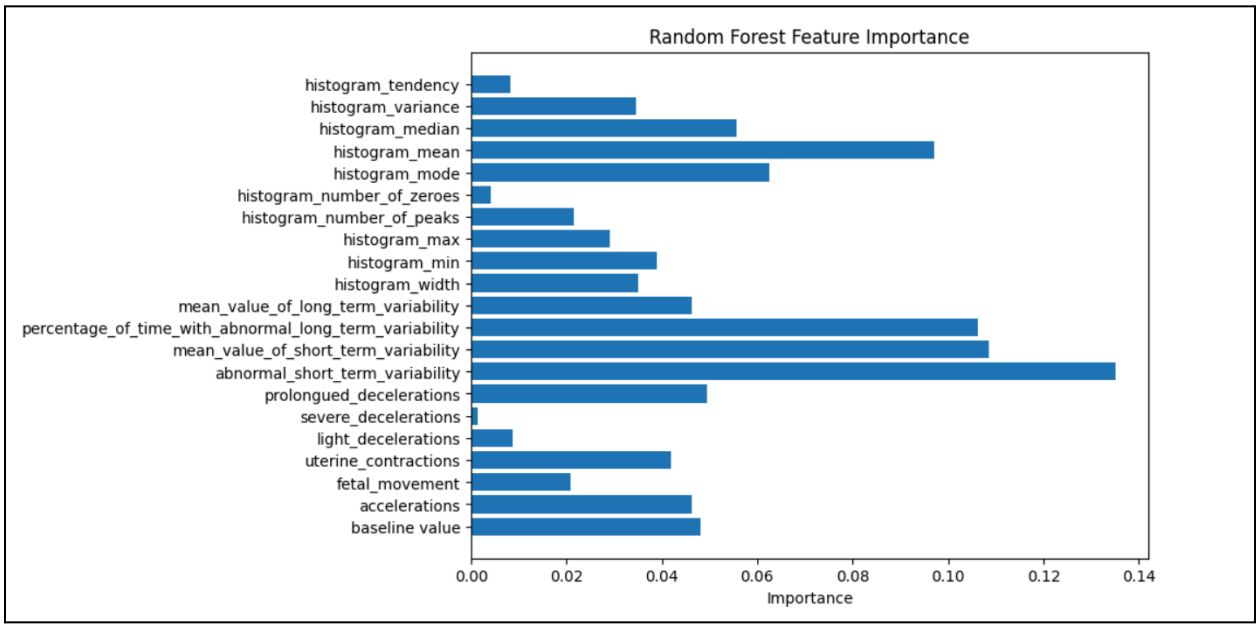
OUTPUT:

```
   baseline value  accelerations  fetal_movement  uterine_contractions  \
0           120.0          0.000             0.0                 0.000
1           132.0          0.006             0.0                 0.006
2           133.0          0.003             0.0                 0.008
3           134.0          0.003             0.0                 0.008
4           132.0          0.007             0.0                 0.008

   light_decelerations  severe_decelerations  prolongued_decelerations  \
0                0.000                   0.0                       0.0
1                0.003                   0.0                       0.0
2                0.003                   0.0                       0.0
3                0.003                   0.0                       0.0
4                0.000                   0.0                       0.0

   abnormal_short_term_variability  mean_value_of_short_term_variability  \
0                             73.0                                   0.5
1                             17.0                                   2.1
2                             16.0                                   2.1
3                             16.0                                   2.4
4                             16.0                                   2.4

   percentage_of_time_with_abnormal_long_term_variability  ...  histogram_min  \
0                                                    43.0  ...           62.0
1                                                     0.0  ...           68.0
2                                                     0.0  ...           68.0
3                                                     0.0  ...           53.0
4                                                     0.0  ...           53.0

   histogram_max  histogram_number_of_peaks  histogram_number_of_zeroes  \
0          126.0                        2.0                         0.0
1          198.0                        6.0                         1.0
2          198.0                        5.0                         1.0
3          170.0                       11.0                         0.0
4          170.0                        9.0                         0.0

   histogram_mode  histogram_mean  histogram_median  histogram_variance  \
0           120.0           137.0             121.0                73.0
1           141.0           136.0             140.0                12.0
2           141.0           135.0             138.0                13.0
3           137.0           134.0             137.0                13.0
4           137.0           136.0             138.0                11.0

   histogram_tendency  fetal_health
0                 1.0           2.0
1                 0.0           1.0
2                 0.0           1.0
3                 1.0           1.0
4                 1.0           1.0
```

```
[5 rows x 22 columns]
(2126, 22)
fetal_health
1.0    1655
2.0     295
3.0     176
Name: count, dtype: int64

--- Decision Tree Classifier ---
Accuracy: 0.9225352112676056
Confusion Matrix:
 [[314  17   2]
 [ 12  52   0]
 [  2   0  27]]
Classification Report:
               precision    recall  f1-score   support

         1.0       0.96      0.94      0.95       333
         2.0       0.75      0.81      0.78        64
         3.0       0.93      0.93      0.93        29

    accuracy                           0.92       426
   macro avg       0.88      0.90      0.89       426
weighted avg       0.92      0.92      0.92       426
```



```
--- Random Forest Classifier ---
Accuracy: 0.9460093896713615
Confusion Matrix:
 [[327   5   1]
 [ 13  49   2]
 [  1   1  27]]
Classification Report:
               precision    recall  f1-score   support

         1.0       0.96      0.98      0.97       333
         2.0       0.89      0.77      0.82        64
         3.0       0.90      0.93      0.92        29

    accuracy                           0.95       426
   macro avg       0.92      0.89      0.90       426
weighted avg       0.94      0.95      0.94       426
```



Conclusion

This experiment successfully demonstrates the application of Decision Tree and Random Forest classifiers on a real-world fetal health dataset. The Random Forest model performs better due to its ensemble approach, making it more reliable for medical classification tasks where accuracy and robustness are critical.